Closing NYC's College Readiness Gap:

A Dashboard and Early Warning Model for Targeted Student Support

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INTRODUCTION

While NYC high school graduation rates have improved from 68.4% in 2014 to 83% in 2024 citywide (Equity NYC, 2025), there is a "Readiness Gap" in the quality of these exits. Specifically, the “Readiness Gap” is made up of students who are graduating from high school despite not being ready to pursue college or a career. Systemic barriers compound this challenge: students in temporary housing and those from schools with a high Economic Need Index (ENI) (which accounts for public assistance and local poverty levels) may be "passed through" to graduation without the academic rigor or stability required to be deemed college-ready, creating a "leaky pipeline" before they even step onto a campus.

Project Vision & Mission

Our project envisions an NYC educational system that can adequately prepare students to succeed beyond high school. By making the "Readiness Gap" visible and quantifiable at the demographic/race level subgroup level, we aim to redefine school quality to move beyond mere graduation rates toward genuine student preparedness. The long-term goal is to close the intergenerational wealth gap by ensuring that the city’s most vulnerable graduates not only reach the graduation stage but also possess the College & Career Readiness necessary to succeed in post-secondary opportunities.

The mission is to provide an evidence-based roadmap that connects high school systemic conditions to actual readiness outcomes through a dashboard and a predictive model. The model will help understand the magnitude and direction of factors contributing to College & Career Readiness (CCR), while the dashboard will serve to explore environmental effects on subgroups. Furthermore, we aim to look for positive outliers– schools that are defying negative factors and providing students with adequate College & Career Readiness. By bridging this data gap, the tool empowers educators to treat college readiness, not just graduation, as the primary high-school-level responsibility.

Project Solution & Impact

Subsequently, we propose a two-part analytical framework designed to establish College & Career Readiness, not just Graduation Rate, as a driving metric for high school success in New York City:

- **The Readiness Dashboard:** Provides a granular view of CCR outcomes disaggregated by demographic subgroups and school contexts. By visualizing how factors like economic need, housing instability, and attendance correlate with readiness, it reframes school quality around postsecondary preparation rather than simple diploma attainment.
- **The Risk Model:** The Model quantifies how systemic and geographic variables influence school-level readiness. Instead of labeling schools as "failing," it establishes evidence-based benchmarks to:
    - Pinpoint the structural factors most strongly associated with CCR.
    - Identify demographic subgroups facing the greatest readiness gaps.
    - Detect "positive outliers"—schools achieving high readiness outcomes despite significant structural disadvantages.

  This emphasis on learning highlights replicable success stories in challenging environments.

The immediate impact will be a realignment towards both CCR and Graduation Rate. By identifying specific contributing factors, decision-makers can move away from broad mandates and toward precision resource allocation. This enables principals to determine the "Readiness Gap" within their own buildings, allowing them to redirect funding toward academic interventions and stability supports for the specific subgroups where the model predicts the highest risk of "unprepared graduation."

Potential Stakeholders

Specifically with our proposal, we envision our primary stakeholders to be the NYC Public School District Leaders and College Access Non-Profits (e.g., College Bridge coaches). These stakeholders need data to determine where to allocate "academic interventionists" and transition coaches who support students while they are still in the building.

Additionally, it is worth noting that the civic education domain as a whole has an array of stakeholders. This includes students, teachers/educators, parents/guardians of students, school administration, and school staff. An in-depth overview is provided in the section on *Stakeholder Conversations & Community Needs*.

CIVIC FOCUS ISSUE

Problem Background

While New York City has successfully boosted graduation rates, these gains haven't guaranteed equitable postsecondary readiness. Significant gaps in College and Career Readiness (CCR) persist, particularly in schools facing high economic need and housing instability. Data from the 2024 School Quality Reports confirms that readiness varies wildly across demographics, challenging the assumption that a diploma inherently equals preparation for college or the workforce (NYC Department of Education, 2025). Ultimately, graduation rates alone are an incomplete measure of school quality and long-term student opportunity.

Furthermore, issues within the system fall hard on educators, who are faced with external policy pressures and reduced curriculum autonomy, which limits teachers' ability to tailor their material to students (Bruxelles, 2025). These challenges are what teachers are recognizing on the ground, with many of them only further propagated by external factors of housing and economic instability.

Root Causes & Prior Interventions

The persistence of this issue is rooted in structural inequalities such as housing access and economic stability. Historically, the NYC DOE has prioritized "on-time graduation" as a metric of success, leading to interventions like credit recovery, which address the immediate requirement of a diploma but fail to address the instability of the learning environment. Prior interventions largely focused on "completion" rather than the academic and social-emotional rigor required for readiness. These solutions are limited because they treat readiness as an individual academic failing rather than a manifestation of systemic neglect during the high school years.

Current interventions appear to react to the symptoms of these systemic issues, rather than deeper problem-solving. Educators in the classrooms have pushed back against the symptoms by shifting their approach with students. This includes developing deep knowledge of students, exposing them to diverse texts, fostering transferable skills, and incorporating flexible scaffolding to support their transitions (Bruxelles, 2025).

Community needs

Based on our analysis, our project focuses on these three critical community needs:

1. **Systemic Accountability for Long-term Outcomes**: The need for school leaders and policymakers to be evaluated not just on graduation, but on the readiness and academic survival of their graduates.

2. **Predictive Support for Housing-Unstable Youth and at-risk student subgroups:** The need for students in temporary housing to receive intensive academic bridging support *before* they exit the system.

3. **Cross-Institutional Data Transparency:** The need for a tool that can show the impact of factors on CCR, allowing communities and schools to assess how external stressors can affect students.

These key needs can help re-evaluate the needs of students and other educational stakeholders, as it looks beyond the straightforward push for higher graduation rates. The problems facing the education domain are much more nuanced, which calls for specific, targeted help rather than a one-size-fits-all solution.

CRITICAL RESEARCH QUESTION & WORKING HYPOTHESIS

Our research played a large role in determining both the factors and outcomes that we chose to focus on. The Economic Need Index (ENI) measures a variety of factors in students, including government assistance and non-English household languages. Capturing this rate alongside the percentage of students in temporary housing allows us to cast a wider net of students inside the Readiness-Gap.

Critical Research Question (CRQ)

To what extent do a high school's Economic Need Index (ENI) and Percent of Students in Temporary Housing predicts its College and Career Readiness (CCR), how do these environmental stressors impact specific demographic subgroups within the same schools,and which schools serve as "positive outliers" by defying these systemic predictors?

Working Hypothesis & Theory of the Problem

The readiness gap is not a failure of individual student effort, but rather a delayed systemic failure rooted in the high school environment. While current policies focus on getting students through to graduation or helping them enroll in college (i.e., Financial Aid), these solutions are responded to differently by students and do not address the systemic issue of poverty and displacement experienced by some high schoolers.

Therefore, we hypothesize that high concentrations of housing instability (school with temp_housing above "threshold") and Economic Need Index (threshold) create an environment of "Survival Learning" over "Mastery Learning." This prevents students from acquiring the academic benchmarks (CCR) needed for post-secondary success, even if they satisfy the basic requirements for a diploma.

PROBLEM ORIENTATION AND ANALYTICAL STANCE

Core Challenge

Many high schools, especially those serving students with high economic need and high rates of temporary housing, can meet graduation requirements without providing all students with equal access to the supports and learning opportunities that build College and Career Readiness (CCR). This creates a gap between the percentage of students who graduate and the percentage who are actually prepared for life after high school. Right now, schools are often judged more by graduation rates than by how ready their students are, which makes this readiness gap harder to see and address. As a result, students from certain demographic groups, particularly those affected by poverty or housing instability, are more likely to graduate without the skills or preparation they need to succeed in college or the workforce. For meaningful improvement to happen, schools and policymakers need to pay closer attention to this gap, particularly College & Career Readiness, and learn from schools that succeed in preparing students for college and careers despite facing economic challenges.

Role of Technology & Data

Data and technology cannot fix the deeper social and economic issues affecting students, such as poverty or housing instability, nor can they replace the relationships, support systems, and resources students need to succeed. However, they can play an important role in helping educators and policymakers better understand where and why gaps in College and Career Readiness (CCR) exist. By analyzing data on the Economic Need Index, temporary housing, graduation rates, and CCR outcomes, this study can reveal patterns that are not always visible when schools are only evaluated by graduation rates. This kind of analysis helps surface inequities between student groups and across schools, showing how certain challenges disproportionately affect readiness even when students earn diplomas. Data can also inform more thoughtful decision-making by identifying schools that are beating the odds and may offer insights into effective practices. While technology should not be treated as a solution on its own, it can support accountability by making readiness gaps visible and harder to ignore, encouraging more targeted and equitable action.

ECOSYSTEM MAP

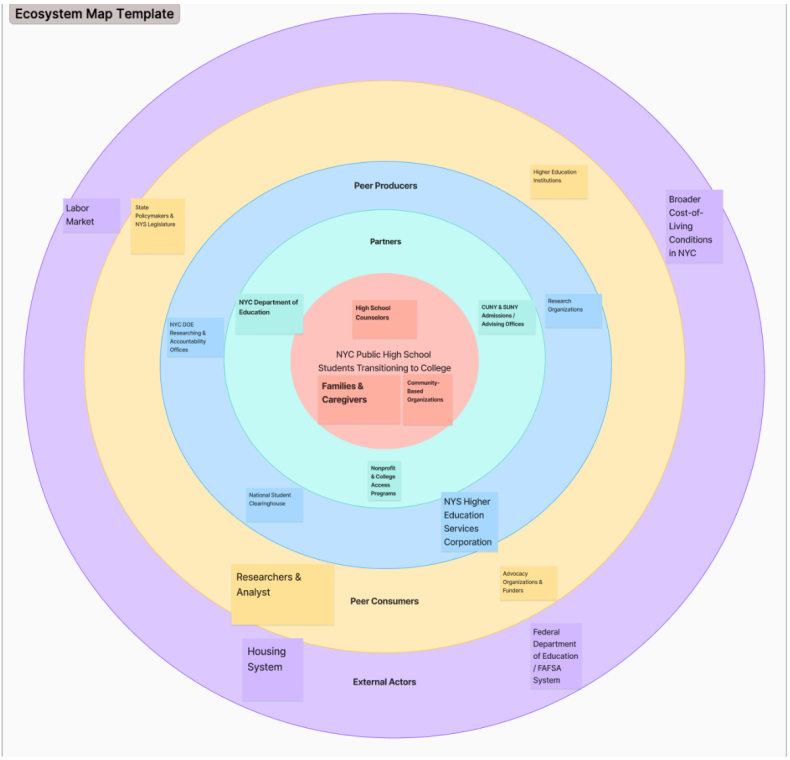
Our ecosystem map aims to visualize the actors and power dynamics at play in the educational civic domain. Centered lies the student, who faces external factors that can influence their college preparedness. Directly next to the student lie the actors that offer direct support, such as parents/guardians, teachers, and counselors. These support persons are often "on the ground," offering direct support to the student through informal means. Finally, the outer circle encompasses all proponents of the educational domain, with large systems and institutions orchestrating policy and funding for all students, not just struggling kids.

Ecosystem Map – Civic Domain of Education

Ecosystem Mapping

The Framework: Five Rings

1. Place your affected community at the center, then identify actors in five concentric rings:
2. **Ring 1: Ecosystem Initiators (Closest to community)**
   a. Who defines the problem? Who has direct relationships with affected communities?
   Examples: CBOs, grassroots advocates, direct service providers
3. **Ring 2: Partners**
   a. Who collaborates to deliver services? Who shares data or resources?
   Examples: Nonprofits, foundations, local agencies, schools
4. **Ring 3: Peer Producers**
   a. Who provides value from outside core relationships? Who generates data?
   Examples: City agencies (DOE, HPD, DOHMH), research institutions, hospitals
5. **Ring 4: Peer Consumers**
   a. Who uses or benefits from ecosystem outputs? Who makes decisions based on data?
   Examples: Policymakers, funders, media, private companies
6. **Ring 5: External Actors (Furthest from community)**
   a. Who shapes context from a distance? What structures constrain or enable action?
   Examples: Federal policies, market forces, tech platforms, cultural narratives



Step-by-Step Process

Round 1: Identify Actors

- Brainstorm all entities connected to your issue
- Write each on a sticky note
- Place in the appropriate ring
- Don't worry about connections yet

Round 2: Draw Connections

Show relationships between actors:

- → Data/information flows (who shares data with whom?)
- $→ Resource flows (who funds what?)
- Power relationships (who has authority?)
- - - - Missing or weak connections

Round 3: Analyze Dynamics

Step back and discuss:

- Power: Who has the most connections? Who controls access?
- Data: Where does your dataset come from? Who collected it?
- Gaps: Whose perspectives are missing? What's invisible in data?
- Positionality: Where do YOU sit in this ecosystem?

STAKEHOLDER CONVERSATIONS & COMMUNITY ENGAGEMENT

A large part of our research falls outside of quantitative data. When it comes to student preparedness and the factors that can impair it, many of the problems just aren't captured by the numbers. For this reason, it was just as important for our team to speak directly with educational stakeholders. Real, anecdotal experiences are crucial for filling in the gaps that the numbers don't capture.

Key Questions for Stakeholders

To explore the intricacies of the domain problem in communities, our team aligned on the idea of giving direct, anecdotal voices a platform. To us, lived experience is a crucial part of understanding the problem, and it is just as important as the quantitative data we will analyze. An important part of tackling this section of our project includes defining what a stakeholder actually is in the context of education. To us, a stakeholder in the education domain includes everyone affected by public education, positively or negatively. This includes, but is not limited to:

- Students
- Teachers/Instructors/Educators
- Parents/Guardians of students
- School Administrators (Principal, School board)
- Community Residents surrounding schools
- Politicians/Policymakers
- School staff (janitors, bus drivers, cafeteria workers)

Given the context of our research question, it was important for us to speak with high school educators in particular, as they fall within the frame of students that we are trying to research. On our part, having thoughtful conversations with stakeholders, we geared our

questions to try to best capture the personal experience, given the role of the stakeholder. Questions include:

Students, Teachers, Community Residents, School Staff

- What/Where is your current role in the context of education? When were you in this role?
- When did you observe the idea of college being introduced to students? To what extent were they introduced to it?
- What do you think your school could've done better in preparing students for college (gaps in college preparedness)?
- Were there any outside factors that made it difficult to focus in school (work, family, mental health, etc.)?

School Administrators, Policymakers, Politicians

- How do you measure student success at your school/district?
- How far did you/your organization keep track of students after going to college? Was this official or personal tracking?
- Did you observe any factors contributing to students who were academically capable of attending college, but dropped out? What were some of those factors, big or small?

Anticipated Insights & Data Limitations

Several holes in our research may go unaddressed, and a large part of this is due to the difficulties surrounding the collection of college student data. There is no straightforward way to link high school student data to larger, public college student data without large assumptions and gaps in consistency. High school graduates also tend to attend college in varying locations, even outside of NY.

For High School students, it will be difficult to get insight into the difficulties of students' lives outside of academics. For this reason, it is crucial to collect real stories from stakeholders in

the education domain. Lived experience can help fill in the gaps in our data and speak on the real leak within the high school to college pipeline.

Planned Influence on the Project

A key component attained from stakeholder conversations will come from the true, lived experiences. Not only can this offer us a guiding hand towards meaningful model variables, but it can also give us a better understanding of our project scope. The factors influencing college preparedness cut across every civic domain in NYC. Economic Need and Housing Stability are just two of many disparate factors contributing to early college drop-out. Thus, overall, stakeholders can help by informing analytical choices such as which variables best serve as proxies for financial coverage or institutional support, and by refining the problem framing, shifting from an individual-level ‘student risk’ indicator to a system-level resource.

ETHICS & EQUITY STATEMENT

Within the civic domain of education, some powerful proponents largely influence policies and decisions. At an institutional level, there are influential policymakers like the NYC Department of Education that determine what educational success looks like in NYC. However, more localized sources of power show up in every public school, which has principals and board members making decisions that influence student resources, counseling, and funding. Further localized, every classroom has a teacher, who is most responsible for the growth and influence of the students within it. These proponents all have an influential stake in the educational domain and play a large role in the way NYC brings up its youth.

The power dynamics therein lie in what the data highlights and what it hides. Students with unstable housing or in need of government assistance are falling through the educational cracks and are largely being hidden under the guise of successful graduation rates. This is covering a large population of students who are experiencing housing instability or attending high-Economic Need Index schools despite high graduation rates within the school district.

Neighborhood changes are reflecting these power dynamics, too. Particularly in Queens and Kings (Brooklyn) counties, gentrifying neighborhoods are seeing an influx of White students from affluent families. These students typically have access to out-of-school resources like enrichment programs, college prep, and social networking that greatly improve their college preparedness, setting them up for future success. Meanwhile, long-standing, low-income communities of color do not have the capital to access these same resources, despite being in the same school district zone. Despite this, aggregated data analysis of school graduation rates and resources effectively hides these struggling students and families, allowing these issues to fall through the cracks (Mordechay & Ayscue, 2022).

With our approach to tackling these issues, our team also had to consider the implications of our findings, especially in regard to the political climate of the current U.S. In highlighting inequities in NYC communities, there is potential to misconstrue the findings of our report and use our targeted system as a means for punitive action, rather than uplifting communities. To acknowledge this, we will work extensively to showcase invisible stories whilst protecting them from being ousted and targeted.

DATA OVERVIEW

1. Data Connection to CRQ

Our proposed data sources, the NYC School Quality Reports (2019–2024), allow us to measure the connection between a school's environmental stress (ENI & percentage of temporary housing specifically) and its academic output and how that affects subgroup populations unevenly. Our primary outcome, or target variable, will be College & Career Readiness (CCR) for the 4-year cohort.

Our primary variables, corresponding to our outcome, are the Economic Need Index and the Percentage of Temp Housing, which serve as our key indicators of environmental stress experienced by specific student subgroups. This allows us to examine the relationship between concentrated environmental stress and its impact on specific demographic groups.

However, we will be inputting various other columns to find other strong predictors of College & Career Readiness.

2. Data Type

We are working with a public administrative dataset providing the NYC Department of Education. This data is appropriate because it captures "the actual outcomes" of students as they navigate high school in different conditions. Having graduation rate and college readiness broken down by demographic subgroup allows us to make the comparison and determine how housing instability and economic need index impact these subgroups, and how that impact translates in Readiness Gap.

3. Data Scope & Scale

Scale: We are working with approximately 500 high schools. Across 4 or 6 years (2019–2024) and multiple demographic subgroups, our "stacked" fact table will consist of roughly 12,000 to 16,000 rows and 40+ columns.

Time Period: *Years of 2024-2025.*

Unit of Analysis: The Subgroup-within-a-School (e.g., Black students at DBN 01M292).

4. **Data Shape & Constraints**

The nature of this data dictates a correlational and predictive rather than causal approach. Because the DOE suppresses small N-counts (privacy protection), we are constrained to analyzing schools with established subgroup populations, potentially excluding very small minorities.

TECHNICAL SPECIFICATIONS, DELIVERABLES & KEY CHALLENGES

I. Problem Framing & Data Sourcing

Component	Finalized Detail
Critical Research Question (CRQ) as it pertains to the Data *To what extent do a high school's Economic Need Index (ENI) and Percent of Students in Temporary Housing predict its College and Career Readiness (CCR), how do these environmental stressors impact specific demographic subgroups within the same schools, and which schools serve as "positive outliers" by defying these systemic predictors?*	
Data Source 1	https://infohub.nyced.org/docs/default-source/default-document-library/202425-hs-sqr-results.xlsx

	Number of Rows: 511 **Number of Columns:** 4 tables **Aggregation:** High school level (some columns broken down per subgroup) **Potential Features to Engineer:** Interactions terms: ENI * subgroup_metrics, Housing_percentage * subgroup_metrics Economic_tiers: ENI tier_1, ENI tier_2 etc
Statistical Test(s)	**Goal:** Test statistically significant differences in College Readiness between subgroups based on ENI and % of temporary housing students **Statistical Test:** ANOVA, Independent T-Tests **Test Details:** Conduct ANOVA for all subgroups. If stat. sig. differences found, we can do Post-Hoc and analysis to determine which groups are higher/lower. From there, we can conduct Independent T-Tests on 2 groups that may be different.
Potential Model(s)- if you don't	**Model 1:** Logistic Regression Model

have a second model, leave it blank	**Y Variable (Target):** College & Career Readiness (CCR) **Evaluation Metric(s):** ENI, % of students in temporary housing **Name of Model 2:** Beta Regression **Y Variable (Target):** College & Career Readiness (CCR) **Evaluation Metric(s):** ENI, Housing Instability Rate, Student Attendance Above 90%

II. Data Limitations & Assumptions

Data Source Name	Potential Limitation(s) / Assumption(s)	Potential Impact on Analysis
School quality report: High school	Our environmental predictors, such as the economic need index and the percentage of temporary housing, are at the high school level, while our outcome, graduation rate, and readiness gap are at the subgroup level. We are assuming "as a baseline" that a 25% ENI in a building affects a Black student the same way it affects a White student in that same building. Our goal will be to	

	determine how those subgroups truly behave given these school conditions relative to each other.	
	Data suppression: If the N count of a metric is less than 15, the rate of that metric is not reported in the data, making it impossible for us to determine the real impact of our features on minorities group, within the same school	This could push our model toward larger schools and may systematically exclude the small ones. Our results might be limited to schools with enough "stable" subgroup populations to be reported.
	Lagging metrics: Our outcome metrics, such as readiness gap, are lagging metrics in the report data, creating a mismatch between the school environment and the outcome. For example, the ENI for a specific high school in the 2024 report does not capture the actual economic need percentage of the graduated class (4-year graduation rate), because this class no longer belongs to the school snap-shot	

	Average bias: the school climate (like ENI and housing percentage) is an average of the entire high school, not only the cohort, which is our outcome. For instance, in 2024, the ENI reported is for all classes from 9th grade to 12th grade in that specific year; the graduation rate is only for the exiting class. However, this metric is pretty consistent over the years.	
	Our readiness gap will capture only students who made it to graduation: Any student who drops out at 11th grade for any reason won't count in our readiness metric, because they did not graduate	

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